Akash Agrawal

Tempe, AZ (Open to relocation) • +1 480-589-7445 • agrawalakash021@gmail.com
linkedin.com/in/akashagrawal021 • akashagrawal.dev

PROFESSIONAL EXPERIENCE

Software Engineer **Feb 2026 - Present**
NewsGenie, Inc. *Mesa, AZ*
Python, FastAPI, LangChain, Claude, ElevenLabs, MongoDB Atlas, Neo4j, Redis, AWS KMS, RAGAS, Datadog, CloudWatch

- Enabled legal-risk checks across 6 countries before publishing by building an AI tool that flags risky phrases and suggests safer wording.
- Saved 1,000+ hours a year by auto-generating 34 AI-adoption courses with Claude Opus 4.5, Claude Sonnet 4.6, and ElevenLabs.
- Built a service that controls how readers and AI agents access publisher content, blocking 96 of 100 test bots in under a second.

AI Engineer **Aug 2024 - Jul 2025**
Arizona State University *Tempe, AZ*
YOLOv8, FairMOT, AWS Lambda, S3, Docker, Terraform, CloudWatch

- Helped civil engineers assess bridge safety by tracking 3,500+ soil particles under vehicle loads at 0.82 mAP.
- Delivered analysis results in 2 hours instead of 40 (95% faster) by building a serverless deep learning pipeline.
- Cut environment setup time 90% and reduced configuration mismatches by standardizing deployments as code.

Software Engineer **Jan 2023 - Nov 2023**
LTIMindtree (Client: Citi Bank) *Remote*
Node.js, Angular, Oracle PL/SQL, REST APIs, SSO, RBAC, Jenkins, Docker, Kubernetes, SonarQube

- Replaced manual configuration changes across financial systems by building automation middleware.
- Sped up financial data processing 35% by optimizing Oracle PL/SQL and integrating REST APIs.
- Made every access to sensitive financial data controlled and traceable with single sign-on, role-based access, and audit logging.

Software Engineer **Aug 2021 - Dec 2022**
LTIMindtree (Client: Avery Dennison) *Remote*
Informatica (ETL, MDM), Python, Oracle PL/SQL, REST APIs, Oracle CRM/CX

- Gave the business a single view of every customer by resolving 10M fragmented records into 2.5M unique customer profiles.
- Improved a legacy pipeline's efficiency 30% through data enrichment.
- Raised data-quality metrics 4x by automating validation tests and removing duplicate records.

PROJECTS

NL2SQL AI Agent **Summer 2025**
Python, RAG, HuggingFace, LangChain, LangGraph, LLM, ChromaDB, SQLite, PostgreSQL

- Cut query turnaround from 3 days to under 1 hour by letting users ask a database questions in plain English.
- Made the agent recover from failures with a self-review step and 3 retry loops, using LangGraph and LangChain.
- Cut hallucinations 60% and reached 0.71 retrieval precision, measured with RAGAS and improved through prompt tuning.

LambdaLens **Fall 2024**
AWS Lambda, OpenCV, Docker, FFmpeg, ResNet-34, MongoDB

- Processed multiple live video streams at once with a serverless analysis pipeline on AWS Lambda.
- Cut cloud costs 50% by optimizing Docker builds, with standardized CloudWatch logs for monitoring.
- Reached 90%+ accuracy on unseen video by running a ResNet-34 model inside the serverless pipeline.

ACHIEVEMENTS

- Ranked 239th internationally out of 10,000 teams in the ICPC 2019.
- Awarded ‘Super Crew’ honor by LTIMindtree for technical excellence and collaboration.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, SQL, C#, C/C++, Bash
AI & LLM: LangChain, LangGraph, RAG, RAGAS, Claude, ElevenLabs, prompt engineering, multi-agent orchestration
Frameworks: FastAPI, React, Node.js, Next.js, GraphQL, RESTful API, Spring Boot, .NET, Angular, Streamlit
Cloud & DevOps: AWS (Lambda, S3, KMS, CloudWatch), Datadog, Docker, Kubernetes, Terraform, Jenkins, CI/CD, GitHub, JIRA
Databases: MySQL, PostgreSQL, Oracle DB, SQLite, MongoDB Atlas, Redis, ChromaDB, pgvector, Neo4j

EDUCATION

M.S. Computer Science | Arizona State University, Tempe, AZ **Dec 2025**
Courses: Statistical Machine Learning, Cloud Computing, Data Processing at Scale, Semantic Web Mining

B.S. Computer Science | Nagpur University, IN **Jul 2021**